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Github: <https://github.com/Soumil-A/HW8SDS.git>

Problem 1

Part A

$$113.723$$

Using linear regression, we predict that a 55-year-old will have a clearance rate of approximately 113.723 mL.

Part B

$$-0.6198159$$

The slope of the regression model is -0.620 , showing that as age increases, creatinine clearance decreases by about 0.620.

Part C

$$11.97972$$

$$1.376035$$

Compared to the model's predictions, the 40-year-old has a creatinine clearance rate 12 mL higher than expected, while the 60-year-old is only about 1.4 mL above expected rate. This shows that the 40-year-old has much healthier kidney function.

Problem 2

Part 1

```
days_since_first_death
      0.183218
days_since_first_death
      3.783183
      2.5 %      97.5 %
0.1623227 0.2041133
      97.5 %      2.5 %
3.395895 4.270180
```

The estimated growth rate is 0.183, with a 95% confidence interval of (0.161, 0.205).

The doubling time is estimated at 3.8 days, with a 95% confidence interval of (3.3, 4.4).

Part 2

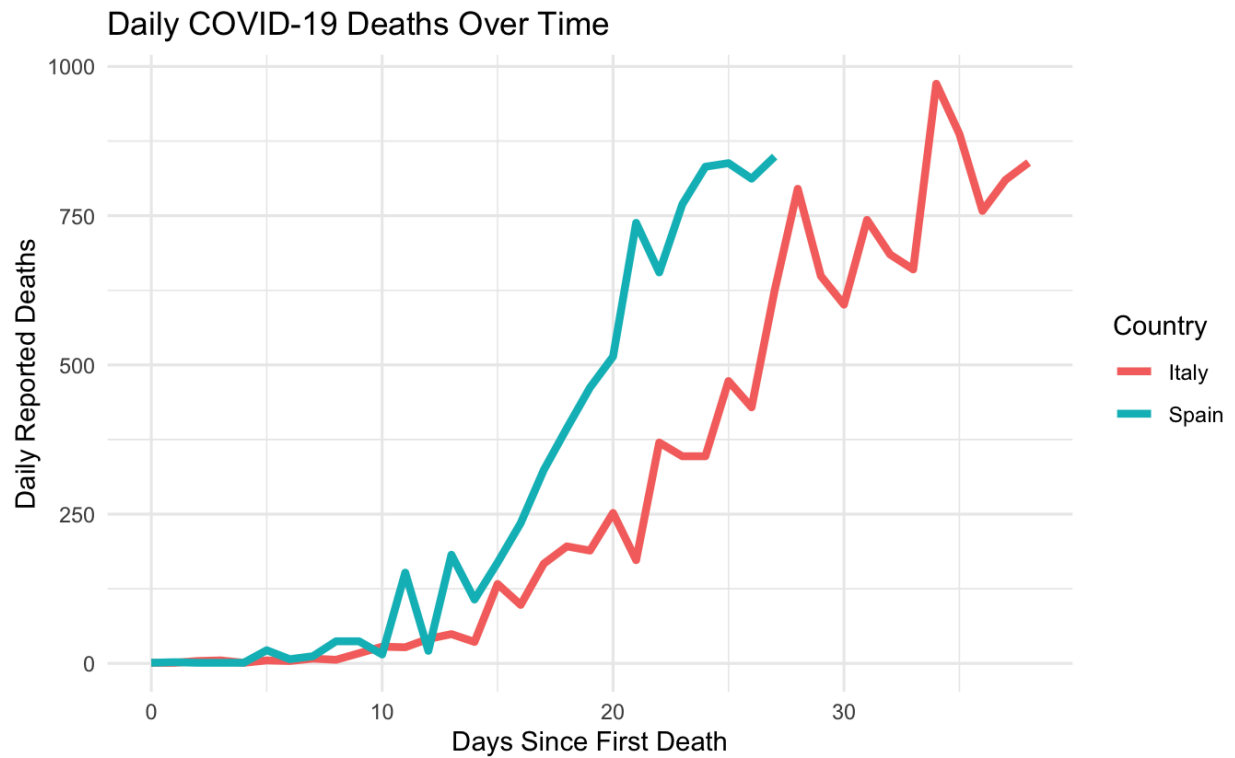
```
days_since_first_death
      0.2762447
days_since_first_death
      2.509178
      2.5 %      97.5 %
0.2369611 0.3155283
      97.5 %      2.5 %
2.196783 2.925151
```

For Spain, the exponential regression model estimates a growth rate of 0.276, with a 95% confidence interval of approximately (0.244, 0.308).

The doubling time is estimated to be about 2.5 days, with a 95% confidence interval of (2.2, 2.9).

Both intervals for both Spain and Italy are statistically significant because they all do not contain 0 in the interval.

Part 3



Problem 3

`log_price` `-1.61858`

`2.5 %` `97.5 %`
`-1.779358` `-1.457798`

The estimated price elasticity of demand for milk is -1.619 . This indicates that for every 1% increase in price, milk sales decrease by about 1.62%.

To find the price elasticity, I took the log of price and sales to use the power-law equation that was given. Then I ran a linear regression in R with $\log(\text{sales})$ and $\log(\text{prices})$ and the slope of that line gave me the elasticity.

The 95% confidence interval for the elasticity is $(-1.779, -1.458)$. This means we are 95% confident that the elasticity falls within that range, and since the interval doesn't include 0, meaning it is statistically significant.